KERALA DIPLOMA CURRICULUM REVISION 2026

Course 4261 | 4 Credits: 4 | Program Core | Source-grounded

Network Switching Technologies

 To provide basics of Network Switching Technologies from a Network

OFFICIAL SOURCE DECLARATION

How this lesson was prepared

The supplied official SITTTR syllabus PDF for Course 4261 is the authoritative source for course information, objectives, course outcomes, CO–PO mapping, modules, topics, activities, assessment structure and references. Explanations, study prompts and Malayalam support are educational elaboration and are not presented as additional official syllabus text.

Source file: 4261.pdf from the uploaded official SITTTR syllabus archive. **Course code and title:** preserved from the source.

COURSE DASHBOARD

Official course information

Course information

Item	Official information
Programme	Diploma in Information Technology
Course code	4261
Course title	Network Switching Technologies
Semester	4 Credits: 4
Course category	Program Core
Periods per week	4 (L:4 T:0 P:0) Periods per semester: 60

Item	Official information
Periods per semester	60
Credits	4
Prerequisite	See the official syllabus PDF
Assessment	Use the official assessment section reproduced below

Modules

4 official module sections detected.

Topic entries

34 source-aligned topic entries retained.

Study mode

Theory and application emphasis.

STUDY METHOD

How to use this lesson

First pass

Read the official source declaration, dashboard and module transcript. Mark unfamiliar terms without changing the official terminology.

Second pass

Use each topic card to build a definition, principle, steps or components, application and exam answer. Attempt the Q&A before opening the answers.

Practical evidence

Where the course is laboratory or workshop based, maintain an observation notebook with procedure, instruments, readings, safety points and conclusion.

Revision pass

Return to the source excerpt, coverage table, activities and model paper. The generated paper is practice material, not an official examination paper.

LEARNING OUTCOMES

Objectives, outcomes and mapping

Course objectives

- To provide basics of Network Switching Technologies from a Network
- Administrator’s point-of-view.
- To provide the basics of Local Area Network, Network Switching and its
- Administration.

Course outcomes

CO	Official outcome	Study level
CO1	15 Applying addressing and subnetting for a simple network. Identify the basic concepts of Internetwork Operating	See official syllabus
CO2	System and management of Internetwork and Network 14 Applying Devices for a simple Network. Identify how to describe, verify, configure troubleshoot	See official syllabus
CO3	switching concepts, port security, VLANs, VLAN 15 Applying Trunking Protocol and Spanning Tree Protocol. Identify the basic concepts of Network Security and	See official syllabus
CO4	14 Applying Network Address Translation for a simple Network.	See official syllabus

CO–PO mapping evidence

Row	Extracted official mapping
CO1	CO1 3
CO2	CO2 3
CO3	CO3 3
CO4	CO4 3

COVERAGE AUDIT

Complete syllabus coverage

Every detected official module is represented below. Each module contains topic cards and an expandable official syllabus transcript to preserve source terminology.

Module coverage

Module	Official heading	Topic entries	Hours
Module 1	subnetting for a simple network.	7	As specified
Module 2	management of Internetwork and Network Devices for a simple Network.	12	As specified
Module 3	port security, VLANs, VLAN Trunking Protocol and Spanning Tree Protocol.	10	As specified
Module 4	Translation for a simple Network.	5	As specified

MODULE 1

subnetting for a simple network.

This module is presented from the official syllabus scope for Network Switching Technologies. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 1 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: subnetting for a simple network.
- Official duration: As specified

- Official topic entries captured: 7
- The full source excerpt is preserved below for coverage checking.

Outline Subnetting and subnet mask.

Outline Subnetting and subnet mask. is an official topic in Module 1 of Network Switching Technologies. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Outline Subnetting and subnet mask. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, outline subnetting and subnet mask. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Outline Subnetting and subnet mask. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-□□ Outline Subnetting and subnet mask. □□□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

Summarize Classless Inter-Domain Routing 1 Understanding Identify Class C Address Subnetting for a simple

Summarize Classless Inter-Domain Routing 1 Understanding Identify Class C Address Subnetting for a simple is an official topic in Module 1 of Network Switching Technologies. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Summarize Classless Inter-Domain Routing 1 Understanding Identify Class C Address Subnetting for a simple using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, summarize classless inter-domain routing 1 understanding identify class c address subnetting for a simple should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Summarize Classless Inter-Domain Routing 1 Understanding Identify Class C Address Subnetting for a simple means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

> **Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-□□ Summarize Classless Inter-Domain Routing 1 Understanding Identify Class C Address Subnetting for a simple □□□□□□□□□□□□ □□□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□, steps, application □□□□□□□ □□□□□□□□□□ □□□□□□□. Technical terms English-□ □□□□□□ □□□□□□□□□□□□.

3 Applying network Identify Class B Address Subnetting for a simple

3 Applying network Identify Class B Address Subnetting for a simple is an official topic in Module 1 of Network Switching Technologies. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3 Applying network Identify Class B Address Subnetting for a simple using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 applying network identify class b address subnetting for a simple should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 3 Applying network Identify Class B Address Subnetting for a simple means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related

but different term.

Malayalam support: Module 1-[illegible] 3 Applying network Identify Class B Address Subnetting for a simple [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

3 Applying network Identify Class A Address Subnetting for a simple

3 Applying network Identify Class A Address Subnetting for a simple is an official topic in Module 1 of Network Switching Technologies. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3 Applying network Identify Class A Address Subnetting for a simple using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 applying network identify class a address subnetting for a simple should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 3 Applying network Identify Class A Address Subnetting for a simple means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related

but different term.

Malayalam support: Module 1-□□ 3 Applying network Identify Class A Address Subnetting for a simple □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

3 Applying network Summarize Variable Length Subnet Mask

3 Applying network Summarize Variable Length Subnet Mask is an official topic in Module 1 of Network Switching Technologies. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3 Applying network Summarize Variable Length Subnet Mask using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 applying network summarize variable length subnet mask should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 3 Applying network Summarize Variable Length Subnet Mask means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related

but different term.

Malayalam support: Module 1-□□ 3 Applying network Summarize Variable Length Subnet Mask □□□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

Design and Implementation for a simple 2 Understanding network.

Design and Implementation for a simple 2 Understanding network. is an official topic in Module 1 of Network Switching Technologies. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Design and Implementation for a simple 2 Understanding network. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, design and implementation for a simple 2 understanding network. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Design and Implementation for a simple 2 Understanding network. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related

but different term.

Malayalam support: Module 1-□□ Design and Implementation for a simple 2 Understanding network. □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

Outline Troubleshooting IP Addressing 2 Understanding Subnetting – Subnetting Steps – Subnet Mask – Classless Inter-Domain Routing – Subnetting Class C Address – Subnetting Class B Address – Subnetting Class A Address Variable Length Subnet Masks – VLSM Design – Implementing VLSM Networks – Troubleshooting IP Addressing Identify the basic concepts of Internetwork Operating System and

Outline Troubleshooting IP Addressing 2 Understanding Subnetting – Subnetting Steps – Subnet Mask – Classless Inter-Domain Routing – Subnetting Class C Address – Subnetting Class B Address – Subnetting Class A Address Variable Length Subnet Masks – VLSM Design – Implementing VLSM Networks – Troubleshooting IP Addressing Identify the basic concepts of Internetwork Operating System and is an official topic in Module 1 of Network Switching Technologies. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Outline Troubleshooting IP Addressing 2 Understanding Subnetting – Subnetting Steps – Subnet Mask – Classless Inter-Domain Routing – Subnetting Class C Address – Subnetting Class B Address – Subnetting Class A Address Variable Length Subnet Masks – VLSM Design – Implementing VLSM Networks – Troubleshooting IP Addressing Identify the basic concepts of Internetwork Operating System and using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, outline troubleshooting ip addressing 2 understanding subnetting – subnetting steps – subnet mask – classless inter-domain routing – subnetting class c address – subnetting class b address – subnetting class a address variable length subnet masks – vlsm design – implementing vlsm networks – troubleshooting ip addressing identify the basic concepts of internetwork operating system and should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Outline Troubleshooting IP Addressing 2 Understanding Subnetting – Subnetting Steps – Subnet Mask – Classless Inter-Domain Routing – Subnetting Class C Address – Subnetting Class B Address – Subnetting Class A Address Variable Length Subnet Masks – VLSM Design – Implementing VLSM Networks – Troubleshooting IP Addressing Identify the basic concepts of Internetwork Operating System and means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-□□ Outline Troubleshooting IP Addressing 2 Understanding Subnetting – Subnetting Steps – Subnet Mask – Classless Inter-Domain Routing – Subnetting Class C Address – Subnetting Class B Address – Subnetting Class A Address Variable Length Subnet Masks – VLSM Design – Implementing VLSM Networks – Troubleshooting IP Addressing Identify the basic concepts of Internetwork Operating System and □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□□.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

```
subnetting for a simple network.
  M1.01      Outline Subnetting and subnet mask.                        1
Understanding
  M1.02      Summarize Classless Inter-Domain Routing                   1
Understanding
             Identify Class C Address Subnetting for a simple
  M1.03                                                                 3
Applying
             network
             Identify Class B Address Subnetting for a simple
  M1.04                                                                 3
Applying
             network
             Identify Class A Address Subnetting for a simple
  M1.05                                                                 3
Applying
             network
             Summarize Variable Length Subnet Mask
  M1.06      Design and Implementation for a simple                     2
Understanding
             network.
  M1.07      Outline Troubleshooting IP Addressing                      2
Understanding
Contents:
Subnetting – Subnetting Steps – Subnet Mask – Classless Inter-Domain Routing –
Subnetting Class C Address – Subnetting Class B Address – Subnetting Class A
Address
```

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 2

management of Internetwork and Network Devices for a simple Network.

This module is presented from the official syllabus scope for Network Switching Technologies. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 2 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: management of Internetwork and Network Devices for a simple Network.
- Official duration: As specified
- Official topic entries captured: 12
- The full source excerpt is preserved below for coverage checking.

1 Understanding (IOS) Illustrate basic IOS Command-Line Interface

1 Understanding (IOS) Illustrate basic IOS Command-Line Interface is an official topic in Module 2 of Network Switching Technologies. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 1 Understanding (IOS) Illustrate basic IOS Command-Line Interface using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 1 understanding (ios) illustrate basic ios command-line interface should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 1 Understanding (IOS) Illustrate basic IOS Command-Line Interface means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related

but different term.

Malayalam support: Module 2-□□ 1 Understanding (IOS) Illustrate basic IOS Command-Line Interface □□□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□□ □□□□□□□□, steps, application □□□□□□ □□□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

2 Understanding (CLI) Commands

2 Understanding (CLI) Commands is an official topic in Module 2 of Network Switching Technologies. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 2 Understanding (CLI) Commands using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 2 understanding (cli) commands should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 2 Understanding (CLI) Commands means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-□□ 2 Understanding (CLI) Commands □□□□□□□□□□□□□ □□□□□□ definition/meaning □□□□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□, steps, application □□□□□□□ □□□□□□□□□□ □□□□□□□. Technical terms English-□ □□□□□□ □□□□□□□□□□□□.

Summarize Administrative Configuration IOS 2 Understanding CLI Commands

Summarize Administrative Configuration IOS 2 Understanding CLI Commands is an official topic in Module 2 of Network Switching Technologies. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Summarize Administrative Configuration IOS 2 Understanding CLI Commands using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, summarize administrative configuration ios 2 understanding cli commands should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Summarize Administrative Configuration IOS 2 Understanding CLI Commands means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related

but different term.

Malayalam support: Module 2-□□ Summarize Administrative Configuration IOS 2 Understanding CLI Commands □□□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□, steps, application □□□□□□□ □□□□□□□□□□ □□□□□□□. Technical terms English-□ □□□□□□ □□□□□□□□□□□□□□.

Outline Router and Switch Interfaces 1 Understanding Summarize Viewing, Saving and Erasing

Outline Router and Switch Interfaces 1 Understanding Summarize Viewing, Saving and Erasing is an official topic in Module 2 of Network Switching Technologies. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Outline Router and Switch Interfaces 1 Understanding Summarize Viewing, Saving and Erasing using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, outline router and switch interfaces 1 understanding summarize viewing, saving and erasing should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Outline Router and Switch Interfaces 1 Understanding Summarize Viewing, Saving and Erasing means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-□□ Outline Router and Switch Interfaces 1 Understanding Summarize Viewing, Saving and Erasing □□□□□□□□□□□□ □□□□□□ definition/meaning □□□□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□, steps, application □□□□□□□ □□□□□□□□□□ □□□□□□□. Technical terms English-□ □□□□□□ □□□□□□□□□□□□□.

1 Understanding Configurations Illustrate the internal components of a Router

1 Understanding Configurations Illustrate the internal components of a Router is an official topic in Module 2 of Network Switching Technologies. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 1 Understanding Configurations Illustrate the internal components of a Router using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 1 understanding configurations illustrate the internal components of a router should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 1 Understanding Configurations Illustrate the internal components of a Router means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

> **Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-□□ 1 Understanding Configurations Illustrate the internal components of a Router □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

1 Understanding and Switch Identify Backing up and Restoring the

1 Understanding and Switch Identify Backing up and Restoring the is an official topic in Module 2 of Network Switching Technologies. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 1 Understanding and Switch Identify Backing up and Restoring the using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 1 understanding and switch identify backing up and restoring the should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 1 Understanding and Switch Identify Backing up and Restoring the means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related

but different term.

Malayalam support: Module 2-[illegible] 1 Understanding and Switch Identify Backing up and Restoring the [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

1 Applying Configuration

1 Applying Configuration is an official topic in Module 2 of Network Switching Technologies. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 1 Applying Configuration using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 1 applying configuration should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 1 Applying Configuration means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-□□ 1 Applying Configuration □□□□□□□□□□□□□ □□□□□□ definition/meaning □□□□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□, steps, application □□□□□□□ □□□□□□□□□□ □□□□□□□. Technical terms English-□ □□□□□□ □□□□□□□□□□□□.

Outline the Configuration of DHCP

Outline the Configuration of DHCP is an official topic in Module 2 of Network Switching Technologies. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Outline the Configuration of DHCP using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, outline the configuration of dhcp should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Outline the Configuration of DHCP means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-[illegible] Outline the Configuration of DHCP [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

Outline usage of Telnet

Outline usage of Telnet is an official topic in Module 2 of Network Switching Technologies. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Outline usage of Telnet using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, outline usage of telnet should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Outline usage of Telnet means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-□□ Outline usage of Telnet □□□□□□□□□□□□ □□□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□, steps, application □□□□□□□ □□□□□□□□□□ □□□□□□□. Technical terms English-□ □□□□□□ □□□□□□□□□□□□□.

Outline how to resolve Hostnames 1 Understanding Summarize how to check network connectivity

Outline how to resolve Hostnames 1 Understanding Summarize how to check network connectivity is an official topic in Module 2 of Network Switching Technologies. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Outline how to resolve Hostnames 1 Understanding Summarize how to check network connectivity using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, outline how to resolve hostnames 1 understanding summarize how to check network connectivity should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Outline how to resolve Hostnames 1 Understanding Summarize how to check network connectivity means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

> **Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-□□ Outline how to resolve Hostnames 1 Understanding Summarize how to check network connectivity □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

1 Understanding and its troubleshooting

1 Understanding and its troubleshooting is an official topic in Module 2 of Network Switching Technologies. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 1 Understanding and its troubleshooting using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 1 understanding and its troubleshooting should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 1 Understanding and its troubleshooting means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-[illegible] 1 Understanding and its troubleshooting [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

Identify Backing up and restoring IOS 1 Applying IOS Basics – Command-Line Interface (CLI) (enable, disable, logout, interface, configure terminal commands) – Administrative Configurations (Hostname, Banners, Passwords Interface Descriptions commands) – Router and Switch Interfaces (Bringing up an interface, configuring an IP address, serial interface command) – Viewing, Saving and Erasing Configurations Internal components of a router and switch – Backing up and Restoring the Configuration – Configuration of DHCP – Using Telnet – Resolving Hostname – Checking Network Connectivity and Troubleshooting – Backing up and restoring IOS Identify how to describe, verify, configure troubleshoot switching concepts,

Identify Backing up and restoring IOS 1 Applying IOS Basics – Command-Line Interface (CLI) (enable, disable, logout, interface, configure terminal commands) – Administrative Configurations (Hostname, Banners, Passwords Interface Descriptions commands) – Router and Switch Interfaces (Bringing up an interface, configuring an IP address, serial interface command) – Viewing, Saving and Erasing Configurations Internal components of a router and switch – Backing up and Restoring the Configuration – Configuration of DHCP – Using Telnet – Resolving Hostname – Checking Network Connectivity and Troubleshooting – Backing up and restoring IOS Identify how to describe, verify, configure troubleshoot switching concepts, is an official topic in Module 2 of Network Switching Technologies. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Identify Backing up and restoring IOS 1 Applying IOS Basics – Command-Line Interface (CLI) (enable, disable, logout, interface, configure terminal commands) – Administrative Configurations (Hostname, Banners, Passwords Interface Descriptions commands) – Router and Switch Interfaces (Bringing up an interface, configuring an IP address, serial interface command) – Viewing, Saving and Erasing Configurations Internal components of a router and switch – Backing up and Restoring the Configuration – Configuration of DHCP – Using Telnet – Resolving Hostname – Checking Network Connectivity and Troubleshooting – Backing up and restoring IOS Identify how to describe, verify, configure troubleshoot switching concepts, using standard technical terminology.

- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, identify backing up and restoring ios 1 applying ios basics – command-line interface (cli) (enable, disable, logout, interface, configure terminal commands) – administrative configurations (hostname, banners, passwords interface descriptions commands) – router and switch interfaces (bringing up an interface, configuring an ip address, serial interface command) – viewing, saving and erasing configurations internal components of a router and switch – backing up and restoring the configuration – configuration of dhcp – using telnet – resolving hostname – checking network connectivity and troubleshooting – backing up and restoring ios identify how to describe, verify, configure troubleshoot switching concepts, should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Identify Backing up and restoring IOS 1 Applying IOS Basics – Command-Line Interface (CLI) (enable, disable, logout, interface, configure terminal commands) – Administrative Configurations (Hostname, Banners, Passwords Interface Descriptions commands) – Router and Switch Interfaces (Bringing up an interface, configuring an IP address, serial interface command) – Viewing, Saving and Erasing Configurations Internal components of a router and switch – Backing up and Restoring the Configuration – Configuration of DHCP – Using Telnet – Resolving Hostname – Checking Network Connectivity and Troubleshooting – Backing up and restoring IOS Identify how to describe, verify, configure troubleshoot switching concepts, means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-□□ Identify Backing up and restoring IOS 1 Applying IOS Basics – Command-Line Interface (CLI) (enable, disable, logout, interface, configure terminal commands) – Administrative Configurations (Hostname, Banners, Passwords Interface Descriptions commands) – Router and

Switch Interfaces (Bringing up an interface, configuring an IP address, serial interface command) – Viewing, Saving and Erasing Configurations Internal components of a router and switch – Backing up and Restoring the Configuration – Configuration of DHCP – Using Telnet – Resolving Hostname – Checking Network Connectivity and Troubleshooting – Backing up and restoring IOS Identify how to describe, verify, configure troubleshoot switching concepts, □□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

```
management of Internetwork and Network Devices for a simple Network.
             Summarize    Internetwork   Operating    System
  M2.01                                                              1
Understanding
             (IOS)
             Illustrate basic IOS Command-Line Interface
  M2.02                                                              2
Understanding
             (CLI) Commands
  M2.03      Summarize Administrative Configuration IOS              2
Understanding
            CLI Commands
  M2.04     Outline Router and Switch Interfaces                   1
Understanding
            Summarize     Viewing,       Saving    and   Erasing
  M2.05                                                            1
Understanding
            Configurations
            Illustrate the internal components of a Router
  M2.06                                                            1
Understanding
            and Switch
            Identify Backing        up    and     Restoring   the
  M2.07                                                            1
Applying
            Configuration
  M2.08     Outline the Configuration of DHCP                      1
```

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 3

port security, VLANs, VLAN Trunking Protocol and Spanning Tree Protocol.

This module is presented from the official syllabus scope for Network Switching Technologies. Use the topic cards for learning and the source transcript for exact coverage.

As specified	**CO-linked**
official hours	see outcomes

Learning goals

- Explain the official Module 3 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: port security, VLANs, VLAN Trunking Protocol and Spanning Tree Protocol.
- Official duration: As specified
- Official topic entries captured: 10
- The full source excerpt is preserved below for coverage checking.

Summarize Switching Services

Summarize Switching Services is an official topic in Module 3 of Network Switching Technologies. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Summarize Switching Services using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, summarize switching services should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Summarize Switching Services means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-[illegible] Summarize Switching Services [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

Outline Port Security 1 Understanding Identify how to do basic configurations of a 2

Outline Port Security 1 Understanding Identify how to do basic configurations of a 2 is an official topic in Module 3 of Network Switching Technologies. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Outline Port Security 1 Understanding Identify how to do basic configurations of a 2 using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, outline port security 1 understanding identify how to do basic configurations of a 2 should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Outline Port Security 1 Understanding Identify how to do basic configurations of a 2 means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

> **Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-□□ Outline Port Security 1 Understanding Identify how to do basic configurations of a 2 □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

Applying Switch

Applying Switch is an official topic in Module 3 of Network Switching Technologies. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Applying Switch using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, applying switch should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Applying Switch means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-[illegible] Applying Switch [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

Summarize VLAN and how to identify VLAN

Summarize VLAN and how to identify VLAN is an official topic in Module 3 of Network Switching Technologies. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Summarize VLAN and how to identify VLAN using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, summarize vlan and how to identify vlan should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Summarize VLAN and how to identify VLAN means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-[illegible] Summarize VLAN and how to identify VLAN [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

Outline Routing between VLANs

Outline Routing between VLANs is an official topic in Module 3 of Network Switching Technologies. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Outline Routing between VLANs using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, outline routing between vlans should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Outline Routing between VLANs means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-[illegible] Outline Routing between VLANs [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

Identify how to configure a VLAN 2 Applying Summarize VLAN Trunking Protocol and its 1

Identify how to configure a VLAN 2 Applying Summarize VLAN Trunking Protocol and its 1 is an official topic in Module 3 of Network Switching Technologies. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Identify how to configure a VLAN 2 Applying Summarize VLAN Trunking Protocol and its 1 using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, identify how to configure a vlan 2 applying summarize vlan trunking protocol and its 1 should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Identify how to configure a VLAN 2 Applying Summarize VLAN Trunking Protocol and its 1 means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

> **Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-□□ Identify how to configure a VLAN 2 Applying Summarize VLAN Trunking Protocol and its 1 □□□□□□□□□□□□□ □□□□□□ definition/meaning □□□□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□, steps, application □□□□□□□ □□□□□□□□□□ □□□□□□□. Technical terms English-□ □□□□□□ □□□□□□□□□□□□□.

Understanding modes of operation

Understanding modes of operation is an official topic in Module 3 of Network Switching Technologies. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Understanding modes of operation using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, understanding modes of operation should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Understanding modes of operation means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-□□ Understanding modes of operation □□□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

Identify how to configure VTP 2 Applying Outline Spanning Tree Protocols with terms, 2

Identify how to configure VTP 2 Applying Outline Spanning Tree Protocols with terms, 2 is an official topic in Module 3 of Network Switching Technologies. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Identify how to configure VTP 2 Applying Outline Spanning Tree Protocols with terms, 2 using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, identify how to configure vtp 2 applying outline spanning tree protocols with terms, 2 should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Identify how to configure VTP 2 Applying Outline Spanning Tree Protocols with terms, 2 means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-□□ Identify how to configure VTP 2 Applying Outline Spanning Tree Protocols with terms, 2 □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

Understanding operations and types

Understanding operations and types is an official topic in Module 3 of Network Switching Technologies. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Understanding operations and types using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, understanding operations and types should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Understanding operations and types means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-□□ Understanding operations and types □□□□□□□□□□□□□□ □□□□□□ definition/meaning □□□□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□, steps, application □□□□□□□ □□□□□□□□□□ □□□□□□□. Technical terms English-□ □□□□□□ □□□□□□□□□□□□□.

Summarize EtherChannel 1 Understanding Switching Services – Three switch functions at layer 2 (Address Learning – Forward/Filter decisions, Loop avoidance) – Port Security – Configuring Switches VLAN Basics – Identifying VLANs – Routing between VLANs – Configuring VLANs – Assigning switch ports to VLAN – Configuring Trunk Ports – Defining the allowed VLANs on a Trunk – Changing or Modifying the Trunk Native VLAN. VLAN Trunking Protocol (VTP) – Modes of operation – Configure VTP – Spanning Tree Protocol (STP) – Terms – Operations - Types of STP – EtherChannel Identify the basic concepts of Network Security and Network Address

Summarize EtherChannel 1 Understanding Switching Services – Three switch functions at layer 2 (Address Learning – Forward/Filter decisions, Loop avoidance) – Port Security – Configuring Switches VLAN Basics – Identifying VLANs – Routing between VLANs – Configuring VLANs – Assigning switch ports to VLAN – Configuring Trunk Ports – Defining the allowed VLANs on a Trunk – Changing or Modifying the Trunk Native VLAN. VLAN Trunking Protocol (VTP) – Modes of operation – Configure VTP – Spanning Tree Protocol (STP) – Terms – Operations - Types of STP – EtherChannel Identify the basic concepts of Network Security and Network Address is an official topic in Module 3 of Network Switching Technologies. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Summarize EtherChannel 1 Understanding Switching Services – Three switch functions at layer 2 (Address Learning – Forward/Filter decisions, Loop avoidance) – Port Security – Configuring Switches VLAN Basics – Identifying VLANs – Routing between VLANs – Configuring VLANs – Assigning switch ports to VLAN – Configuring Trunk Ports – Defining the allowed VLANs on a Trunk – Changing or Modifying the Trunk Native VLAN. VLAN Trunking Protocol (VTP) – Modes of operation – Configure VTP – Spanning Tree Protocol (STP) – Terms – Operations - Types of STP – EtherChannel Identify the basic concepts of Network Security and Network Address using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.

- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, summarize etherchannel 1 understanding switching services – three switch functions at layer 2 (address learning – forward/filter decisions, loop avoidance) – port security – configuring switches vlan basics – identifying vlans – routing between vlans – configuring vlans – assigning switch ports to vlan – configuring trunk ports – defining the allowed vlans on a trunk – changing or modifying the trunk native vlan. vlan trunking protocol (vtp) – modes of operation – configure vtp – spanning tree protocol (stp) – terms – operations - types of stp – etherchannel identify the basic concepts of network security and network address should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Summarize EtherChannel 1 Understanding Switching Services – Three switch functions at layer 2 (Address Learning – Forward/Filter decisions, Loop avoidance) – Port Security – Configuring Switches VLAN Basics – Identifying VLANs – Routing between VLANs – Configuring VLANs – Assigning switch ports to VLAN – Configuring Trunk Ports – Defining the allowed VLANs on a Trunk – Changing or Modifying the Trunk Native VLAN. VLAN Trunking Protocol (VTP) – Modes of operation – Configure VTP – Spanning Tree Protocol (STP) – Terms – Operations - Types of STP – EtherChannel Identify the basic concepts of Network Security and Network Address means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-□□ Summarize EtherChannel 1 Understanding Switching Services – Three switch functions at layer 2 (Address Learning – Forward/Filter decisions, Loop avoidance) – Port Security – Configuring Switches VLAN Basics – Identifying VLANs – Routing between VLANs – Configuring VLANs – Assigning switch ports to VLAN – Configuring Trunk Ports – Defining the allowed VLANs on a Trunk – Changing or Modifying the Trunk Native VLAN. VLAN Trunking Protocol (VTP) – Modes of operation – Configure VTP – Spanning Tree Protocol (STP) – Terms – Operations - Types of STP – EtherChannel Identify the basic concepts of

Network Security and Network Address □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

```
port security, VLANs, VLAN Trunking Protocol and Spanning Tree Protocol.
  M3.01     Summarize Switching Services                            2
Understanding
  M3.02     Outline Port Security                                   1
Understanding
            Identify how to do basic configurations of a            2
  M3.03
Applying
            Switch
  M3.04     Summarize VLAN and how to identify VLAN                 1
Understanding
  M3.05     Outline Routing between VLANs                           1
Understanding
  M3.06     Identify how to configure a VLAN                        2
Applying
            Summarize VLAN Trunking Protocol and its                1
  M3.07
Understanding
            modes of operation
  M3.08      Identify how to configure VTP                        2
Applying
             Outline Spanning Tree Protocols with terms,          2
  M3.09
Understanding
             operations and types
  M3.10      Summarize EtherChannel                               1
Understanding
```

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 4

Translation for a simple Network.

This module is presented from the official syllabus scope for Network Switching Technologies. Use the topic cards for learning and the source transcript for exact coverage.

As specified	**CO-linked**
official hours	see outcomes

Learning goals

- Explain the official Module 4 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Translation for a simple Network.
- Official duration: As specified
- Official topic entries captured: 5
- The full source excerpt is preserved below for coverage checking.

Outline Perimeter Router, Firewall and Internal 2 Understanding Router

Outline Perimeter Router, Firewall and Internal 2 Understanding Router is an official topic in Module 4 of Network Switching Technologies. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Outline Perimeter Router, Firewall and Internal 2 Understanding Router using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, outline perimeter router, firewall and internal 2 understanding router should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Outline Perimeter Router, Firewall and Internal 2 Understanding Router means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related

but different term.

Malayalam support: Module 4-□□ Outline Perimeter Router, Firewall and Internal 2 Understanding Router □□□□□□□□□□□□□ □□□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□, steps, application □□□□□□□ □□□□□□□□□□ □□□□□□□. Technical terms English-□ □□□□□□ □□□□□□□□□□□□.

Summarize Standard and Extended Access Lists

Summarize Standard and Extended Access Lists is an official topic in Module 4 of Network Switching Technologies. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Summarize Standard and Extended Access Lists using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, summarize standard and extended access lists should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Summarize Standard and Extended Access Lists means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-□□ Summarize Standard and Extended Access Lists □□□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

Illustrate usage and types of NAT

Illustrate usage and types of NAT is an official topic in Module 4 of Network Switching Technologies. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Illustrate usage and types of NAT using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, illustrate usage and types of nat should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Illustrate usage and types of NAT means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-[illegible] Illustrate usage and types of NAT [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

Outline Working of NAT 3 Understanding Identify the configuration of static and dynamic

Outline Working of NAT 3 Understanding Identify the configuration of static and dynamic is an official topic in Module 4 of Network Switching Technologies. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Outline Working of NAT 3 Understanding Identify the configuration of static and dynamic using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, outline working of nat 3 understanding identify the configuration of static and dynamic should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Outline Working of NAT 3 Understanding Identify the configuration of static and dynamic means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

> **Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-□□ Outline Working of NAT 3 Understanding Identify the configuration of static and dynamic □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□, steps, application □□□□□□ □□□□□□□□□□ □□□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□□.

3 Applying NAT LAN Security - Perimeter Router, Firewall and Internal Router – Access Lists (Standard and Extended) Network Address Translation (NAT) – Use of NAT – Types of NAT – Working of NAT – Static NAT Configuration – Dynamic NAT Configuration Text / Reference: T/R Book Title / Author Todd Lammle, CCNA Routing and Switching Complete Study Guide, 2nd Edition, T1 Sybex (Wiley) Publication, 2016 Narasimha Karumanchi, Dr. A. Damodaram and Dr. M. Sreenivasa Rao. Elements of R1 Computer Networking: An Integrated Approach. CareerMonk Publications, 2014. Online Resources: Sl.No Website Link 1 https://www.freeccnastudyguide.com/study-guides/icnd1-100-101/ https://www.thebryantadvantage.com/videos-and-tutorials/ccna-and-ccent- 2 tutorials/ 3 https://www.guru99.com/introduction-ccna.html 4 https://study-ccna.com/ 5 http://www.omnisecu.com/cisco-certified-network-associate-ccna/index.php

3 Applying NAT LAN Security - Perimeter Router, Firewall and Internal Router – Access Lists (Standard and Extended) Network Address Translation (NAT) – Use of NAT – Types of NAT – Working of NAT – Static NAT Configuration – Dynamic NAT Configuration Text / Reference: T/R Book Title / Author Todd Lammle, CCNA Routing and Switching Complete Study Guide, 2nd Edition, T1 Sybex (Wiley) Publication, 2016 Narasimha Karumanchi, Dr. A. Damodaram and Dr. M. Sreenivasa Rao. Elements of R1 Computer Networking: An Integrated Approach. CareerMonk Publications, 2014. Online Resources: Sl.No Website Link 1 https://www.freeccnastudyguide.com/study-guides/icnd1-100-101/ https://www.thebryantadvantage.com/videos-and-tutorials/ccna-and-ccent- 2 tutorials/ 3 https://www.guru99.com/introduction-ccna.html 4 https://study-ccna.com/ 5 http://www.omnisecu.com/cisco-certified-network-associate-ccna/index.php is an official topic in Module 4 of Network Switching Technologies. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3 Applying NAT LAN Security - Perimeter Router, Firewall and Internal Router – Access Lists (Standard and Extended) Network Address Translation (NAT) – Use of NAT – Types of NAT – Working of NAT – Static NAT Configuration – Dynamic NAT Configuration Text / Reference: T/R Book Title /

Author Todd Lammle, CCNA Routing and Switching Complete Study Guide, 2nd Edition, T1 Sybex (Wiley) Publication, 2016 Narasimha Karumanchi, Dr. A. Damodaram and Dr. M. Sreenivasa Rao. Elements of R1 Computer Networking: An Integrated Approach. CareerMonk Publications, 2014. Online Resources: Sl.No Website Link 1 https://www.freeccnastudyguide.com/study-guides/icnd1-100-101/ https://www.thebryantadvantage.com/videos-and-tutorials/ccna-and-ccent- 2 tutorials/ 3 https://www.guru99.com/introduction-ccna.html 4 https://study-ccna.com/ 5 http://www.omnisecu.com/cisco-certified-network-associate-ccna/index.php using standard technical terminology.

- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 applying nat lan security - perimeter router, firewall and internal router – access lists (standard and extended) network address translation (nat) – use of nat – types of nat – working of nat – static nat configuration – dynamic nat configuration text / reference: t/r book title / author todd lammle, ccna routing and switching complete study guide, 2nd edition, t1 sybex (wiley) publication, 2016 narasimha karumanchi, dr. a. damodaram and dr. m. sreenivasa rao. elements of r1 computer networking: an integrated approach. careermonk publications, 2014. online resources: sl.no website link 1 https://www.freeccnastudyguide.com/study-guides/icnd1-100-101/ https://www.thebryantadvantage.com/videos-and-tutorials/ccna-and-ccent- 2 tutorials/ 3 https://www.guru99.com/introduction-ccna.html 4 https://study-ccna.com/ 5 http://www.omnisecu.com/cisco-certified-network-associate-ccna/index.php should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 3 Applying NAT LAN Security - Perimeter Router, Firewall and Internal Router – Access Lists (Standard and Extended) Network Address Translation (NAT) – Use of NAT – Types of NAT – Working of NAT – Static NAT Configuration – Dynamic NAT Configuration Text / Reference: T/R Book Title / Author Todd Lammle, CCNA Routing and Switching Complete Study Guide, 2nd Edition, T1 Sybex (Wiley) Publication, 2016 Narasimha Karumanchi, Dr. A. Damodaram and Dr. M. Sreenivasa Rao. Elements of R1 Computer Networking: An Integrated Approach. CareerMonk Publications, 2014. Online Resources: Sl.No Website Link 1 https://www.freeccnastudyguide.com/study-guides/icnd1-100-101/ https://www.thebryantadvantage.com/videos-and-tutorials/ccna-and-ccent- 2 tutorials/ 3 https://www.guru99.com/introduction-ccna.html 4 https://study-ccna.com/ 5 http://www.omnisecu.com/cisco-certified-network-associate-ccna/index.php means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-□□ 3 Applying NAT LAN Security - Perimeter Router, Firewall and Internal Router – Access Lists (Standard and Extended) Network Address Translation (NAT) – Use of NAT – Types of NAT – Working of NAT – Static NAT Configuration – Dynamic NAT Configuration Text / Reference: T/R Book Title / Author Todd Lammle, CCNA Routing and Switching Complete Study Guide, 2nd Edition, T1 Sybex (Wiley) Publication, 2016 Narasimha Karumanchi, Dr. A. Damodaram and Dr. M. Sreenivasa Rao. Elements of R1 Computer Networking: An Integrated Approach. CareerMonk Publications, 2014. Online Resources: Sl.No Website Link 1 https://www.freeccnastudyguide.com/study-guides/icnd1-100-101/ https://www.thebryantadvantage.com/videos-and-tutorials/ccna-and-ccent- 2 tutorials/ 3 https://www.guru99.com/introduction-ccna.html 4 https://study-ccna.com/ 5 http://www.omnisecu.com/cisco-certified-network-associate-ccna/index.php □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

```
Translation for a simple Network.

  M 4.01     Outline Perimeter Router, Firewall and Internal      2
Understanding
             Router
  M 4.02     Summarize Standard and Extended Access Lists         3
Understanding
  M 4.03     Illustrate usage and types of NAT                    3
Understanding
  M 4.04     Outline Working of NAT                               3
Understanding
             Identify the configuration of static and dynamic
  M 4.05                                                          3
Applying
             NAT

             Series Test – II                                     1

Contents:
LAN Security - Perimeter Router, Firewall and Internal Router – Access Lists
(Standard
and Extended)
Network Address Translation (NAT) – Use of NAT – Types of NAT – Working of NAT
– Static NAT Configuration – Dynamic NAT Configuration

Text / Reference:
```

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

RAPID REFERENCE

Concept and formula bank

Answer structure

```
Definition → Principle → Components/steps → Application → Conclusion
```

Use this sequence for descriptive questions when the topic does not require a numerical formula.

Practical record

```
Aim → Apparatus → Procedure → Observation → Result → Safety
```

Use this sequence for laboratory, workshop and field activities.

RAPID REVISION

Diagram and source-transcript library

Module 1 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 2 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 3 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 4 source and topic cards

Jump to official terminology, topic explanations and study guidance.

OFFICIAL SELF-LEARNING

Activities from the syllabus

Use the extracted official activity wording as the record of required self-learning work.

The official PDF does not expose a separate activities heading in the extracted text; consult the source PDF pages for the exact activity list.

EXAMINATION PREPARATION

Assessment methodology

The following source extract preserves the assessment information detected in the official PDF. Verify mark conversions and institutional updates against the current source before an examination.

```
The official assessment structure is reproduced from the supplied PDF.
```

Practice-paper notice: The model paper below is generated for revision and is not an official examination paper.

ACTIVE RECALL

Module-wise questions and answers

These are practice questions based on official topic coverage, not official examination questions.

Module 1

Define or explain Outline Subnetting and subnet mask.. (3 marks)

Answer: Begin with the standard definition or identity of *Outline Subnetting and subnet mask.*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain Summarize Classless Inter-Domain Routing 1 Understanding Identify Class C Address Subnetting for a simple. (3 marks)

Answer: Begin with the standard definition or identity of *Summarize Classless Inter-Domain Routing 1 Understanding Identify Class C Address Subnetting for a simple*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain 3 Applying network Identify Class B Address Subnetting for a simple. (3 marks)

Answer: Begin with the standard definition or identity of *3 Applying network Identify Class B Address Subnetting for a simple*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain 3 Applying network Identify Class A Address Subnetting for a simple. (3 marks)

Answer: Begin with the standard definition or identity of *3 Applying network Identify Class A Address Subnetting for a simple*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□□□ □□□□□□ definition, □□□□□□□ principle/steps, □□□□□□□ application □□□□ □□□□□□□□□□ □□□□□□□.

Module 2

Define or explain 1 Understanding (IOS) Illustrate basic IOS Command-Line Interface. (3 marks)

Answer: Begin with the standard definition or identity of *1 Understanding (IOS) Illustrate basic IOS Command-Line Interface*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□□□ □□□□□□ definition, □□□□□□□ principle/steps, □□□□□□□ application □□□□ □□□□□□□□□□ □□□□□□□.

Define or explain 2 Understanding (CLI) Commands. (3 marks)

Answer: Begin with the standard definition or identity of *2 Understanding (CLI) Commands*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□□□ □□□□□□ definition, □□□□□□□ principle/steps, □□□□□□□ application □□□□ □□□□□□□□□□ □□□□□□□.

Define or explain Summarize Administrative Configuration IOS 2 Understanding CLI Commands. (3 marks)

Answer: Begin with the standard definition or identity of *Summarize Administrative Configuration IOS 2 Understanding CLI Commands*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain Outline Router and Switch Interfaces 1 Understanding Summarize Viewing, Saving and Erasing. (3 marks)

Answer: Begin with the standard definition or identity of *Outline Router and Switch Interfaces 1 Understanding Summarize Viewing, Saving and Erasing*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Module 3

Define or explain Summarize Switching Services. (3 marks)

Answer: Begin with the standard definition or identity of *Summarize Switching Services*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain Outline Port Security 1 Understanding Identify how to do basic configurations of a 2. (3 marks)

Answer: Begin with the standard definition or identity of *Outline Port Security 1 Understanding Identify how to do basic configurations of a 2*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps,

□□□□□□□ application □□□□□ □□□□□□□□□□ □□□□□□□.

Define or explain Applying Switch. (3 marks)

Answer: Begin with the standard definition or identity of *Applying Switch*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□□ □□□□□□ definition, □□□□□□□ principle/steps, □□□□□□□ application □□□□□ □□□□□□□□□□ □□□□□□□.

Define or explain Summarize VLAN and how to identify VLAN. (3 marks)

Answer: Begin with the standard definition or identity of *Summarize VLAN and how to identify VLAN*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□□ □□□□□□ definition, □□□□□□□ principle/steps, □□□□□□□ application □□□□□ □□□□□□□□□□ □□□□□□□.

Module 4

Define or explain Outline Perimeter Router, Firewall and Internal 2 Understanding Router. (3 marks)

Answer: Begin with the standard definition or identity of *Outline Perimeter Router, Firewall and Internal 2 Understanding Router*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□□ □□□□□□ definition, □□□□□□□ principle/steps, □□□□□□□ application □□□□□ □□□□□□□□□□ □□□□□□□.

Define or explain Summarize Standard and Extended Access Lists. (3 marks)

Answer: Begin with the standard definition or identity of *Summarize Standard and Extended Access Lists*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain Illustrate usage and types of NAT. (3 marks)

Answer: Begin with the standard definition or identity of *Illustrate usage and types of NAT*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain Outline Working of NAT 3 Understanding Identify the configuration of static and dynamic. (3 marks)

Answer: Begin with the standard definition or identity of *Outline Working of NAT 3 Understanding Identify the configuration of static and dynamic*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

PRACTICE ONLY

Practice Model Paper — Not an Official Examination Paper

Attempt one short definition, one structured explanation and one long answer from every module. Use the official assessment pattern reproduced above when selecting time and marks.

Module 1

1-mark definition: State the meaning of **Outline Subnetting and subnet mask.**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 2

1-mark definition: State the meaning of **1 Understanding (IOS) Illustrate basic IOS Command-Line Interface**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 3

1-mark definition: State the meaning of **Summarize Switching Services**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 4

1-mark definition: State the meaning of **Outline Perimeter Router, Firewall and Internal 2 Understanding Router**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

QUICK REVISION

Exam checklist

Before writing

- Read the command word: define, explain, compare, draw, calculate, analyse or design.
- Use the exact course terminology from the source transcript.
- Plan labels, units, sequence and marks before writing.

Before submitting

- Check every module has been revised.
- Check diagrams, tables, formula symbols and units.
- Separate official syllabus information from personal elaboration.

REFERENCE VOCABULARY

Glossary

Study glossary

Term	Meaning in this lesson
Course Outcome	A capability the student should demonstrate after completing the course.
Module	An official syllabus unit with defined topics and hours.
CIE	Continuous Internal Evaluation.
ESE	End Semester Examination.
Source transcript	A preserved extract of the official syllabus used for coverage checking.

SOURCE-SUPPORTED REFERENCES

References and web resources

References listed in the official PDF

References are included in the official source PDF.

Web resources listed in the official PDF

- Sl.No Website Link
- https://www.freeccnastudyguide.com/study-guides/icnd1-100-101/
- https://www.thebryantadvantage.com/videos-and-tutorials/ccna-and-ccent- tutorials/
- https://www.guru99.com/introduction-ccna.html https://study-ccna.com/
- http://www.omnisecu.com/cisco-certified-network-associate-ccna/index.php

IN-PAGE SEARCH

Search this lesson